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Thermal System Modelling Report
ACS133 – Physical Systems

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Summary:

The report discusses and analyses the behaviour and how the temperature changes in room 1 and room 2 as an effect of heat generated by the heat supplier $q(t)$ which is located in room 1.

The heat transfer from the supplier to surrounding in room 1 then to room 2 or to external environment is analysed and modelled considering given and also different and various combinations of wall – Thermal- resistances and room capacitance for each room which will cause changes to model behaviour.

Note:

Although thermal capacitance and thermal resistance are typically associated with distributed bodies in space and not considered lumped elements, in this module we have chosen to treat them as elements in the lumped modelling of thermal systems. As well as the temperature of the room at a certain point of time is assumed to be uniform.

Model Description:

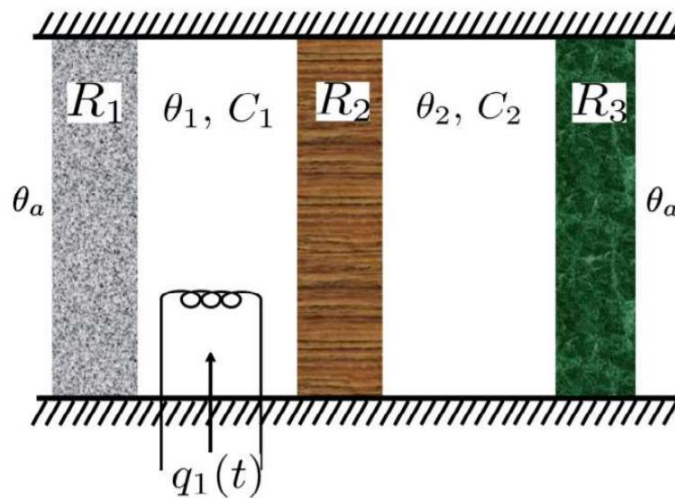


Figure 1

As shown in figure 1, the heat supplier produces heat in room, this heat is modelled as a step response $q_1(t)$. Heat is then transferred to surrounding θ_1 and then to θ_2 via R_2 . Each room has a certain thermal capacity C_1 and C_2 which affects the quantity of heat each room can store.

The relation between θ_1 and $q_1(t)$ can be represented by the following transfer function:

$$\frac{\hat{\theta}_1(s)}{Q_1(s)} = \frac{\frac{1}{C_1} s^2 + s \left(\frac{1}{C_1^2 R_{12}} + \frac{1}{C_1 C_2 R_{23}} \right) + \frac{1}{C_1^2 C_2 R_{12} R_{23}}}{s^3 + s^2 \left(\frac{2}{C_1 R_{12}} + \frac{1}{C_2 R_{23}} \right) + s \left(\frac{1}{C_1^2 R_{12}^2} + \frac{1}{C_1 C_2 R_{12} R_{23}} + \frac{R_2^2 - R_{12} R_{23}}{C_1 C_2 R_{12} R_{23} R_2^2} \right) + \frac{R_2^2 - R_{12} R_{23}}{C_1^2 C_2 R_{12}^2 R_{23} R_2^2}}$$

For the transfer function of $\frac{\theta_1(s)}{Q_1(s)}$ is represented in MATLAB as:

```
num1 = [1/C_1 (1/(R_12*(C_1^2)))+(1/(C_1*C_2*R_23))] 1/(R_23*R_12*C_2*(C_1^2))
den1 = [1 1/(C_2*R_23)+2/(C_1*R_12) (1/((C_1^2)*(R_12^2)))+(1/(C_1*C_2*R_12*R_23))+((R_2^2)-
(R_12*R_23))/(C_1*C_2*(R_12)*R_23*(R_2^2)) ((R_2^2)-(R_12*R_23))/(C_1^2*C_2*(R_12^2)*R_23*(R_2^2))]
Theta1_TF = tf(num1,den1); % transfer function theta1(s)/Q1(s)
```

MATLAB code snippet 1

The relation between θ_2 and $q_1(t)$ can be represented by the following transfer function:

$$\frac{\hat{\Theta}_2(s)}{Q_1(s)} = \frac{\frac{1}{C_1 C_2 R_2}}{s^2 + s \left(\frac{1}{C_1 R_{12}} + \frac{1}{C_2 R_{23}} \right) + \frac{R_2^2 - R_{12} R_{23}}{C_1 C_2 R_{12} R_{23} R_2^2}}$$

For the transfer function of $\frac{\theta_2(s)}{Q_1(s)}$ is represented in MATLAB as:

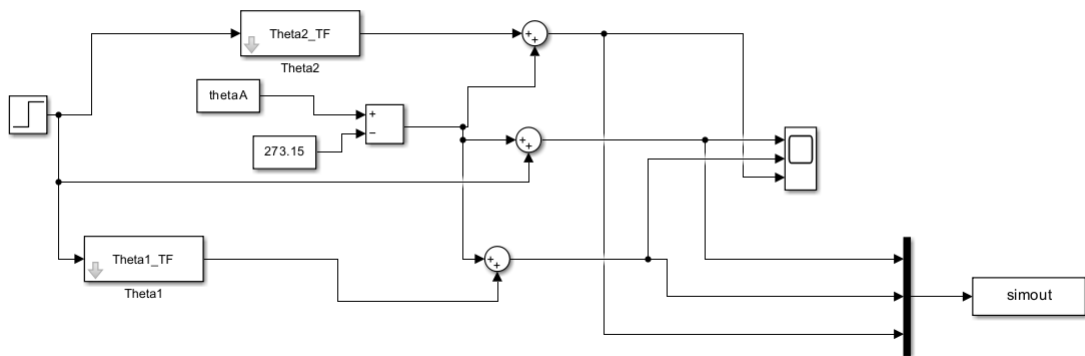
```
num2 = 1/(C_1*C_2*R_2);
den2 = [1 ((1/(C_1*R_12)))+(1/(C_2*R_23))) ((R_2^2)-(R_12*R_23))/(C_1*C_2*R_12*R_23*R_2^2)];
Theta2_TF = tf(num2,den2); % transfer function theta2(s)/Q1(s)
```

MATLAB code snippet 2

Where R_{12} and R_{23} are:

$$R_{12} = \frac{R_1 R_2}{R_1 + R_2}, \quad R_{23} = \frac{R_2 R_3}{R_2 + R_3}.$$

Simulink Overview:



Simulink_1

For the Simulink, the step represents $Q_1(S)$. The step allows the system 10 minutes where $Q_1(S) = 273.15^\circ K$, however in this case $Q_1(S)$ will start from the value of θ_a

Step	
Output a step.	
Main	Signal Attributes
Step time:	
600	
Initial value:	
0	
Final value:	
3	
Sample time:	
0	
☒ Interpret vector parameters as 1-D	
☒ Enable zero-crossing detection	

Simulink_2

Blocks Theta1 and Theta2 are LTI System blocks which takes the value of the transfer functions Theta1_TF and Theta2_TF from MATLAB workspace to the Simulink model. Values then are moved for the step, Theta1 and Theta2 blocks to workspace (simout) and to scope for initial visualisation in case of errors.

Considering given (pre-calculated) values:

Considering these values for our model:

$$C_1 = 50 \frac{J}{K}$$

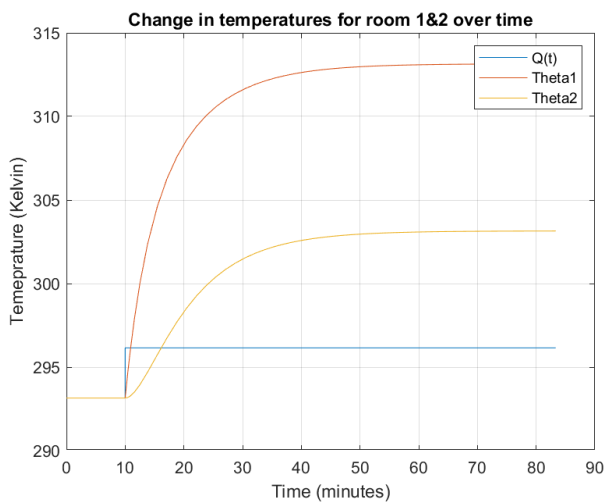
$$C_2 = 60 \frac{J}{K}$$

$$R_1 = 10 \frac{K}{W}$$

$$R_2 = 10 \frac{K}{W}$$

$$R_3 = 10 \frac{K}{W}$$

$$\theta_a = 293.15^\circ K$$



The plot in Figure_2 shows how the system behaviour changes over time. As pre-described, the system will maintain a value of $293.15^\circ K$ for 10 min, since the initial temperature of the system is θ_a . Then the step response is considered, where $Q_1(S) = 296.15$ at $t = 10 \text{ min}$

In this case, the 2 transfer functions are convergent, where Theta1 has higher steady state value than Theta2 due to the effect of wall resistance and heat already stored in room1.

[Table 1](#) below shows detailed comparison between the 2 transfer functions.

Figure 2

Theta1	Theta2
RiseTime: 21.6134	RiseTime: 24.6417
TransientTime: 42.3669	TransientTime: 49.4700
SettlingTime: 36.0668	SettlingTime: 39.5073
SettlingMin: 36.2512	SettlingMin: 27.2595
SettlingMax: 39.9953	SettlingMax: 29.9948
Overshoot: 0	Overshoot: 0
Undershoot: 0	Undershoot: 0
Peak: 313.1453	Peak: 303.1448
PeakTime: 83.3333	PeakTime: 83.3333

Table 1

Note: the 10 min delay is considered in this table and peak values are in °K

For Theta1, it has faster settling time than Theta2 which is due to the effect of $Q(t)$ is affecting Theta1 in a higher rate than Theta2. Theta2 has lower steady state value due to energy lost overcoming the resistance between the 2 rooms.

Analysing model with various values:

For testing and analysing model behaviour with various data, a MATLAB GUI is designed to make it more convenient for user to change parameter values. GUI overview is shown below in Figure 3

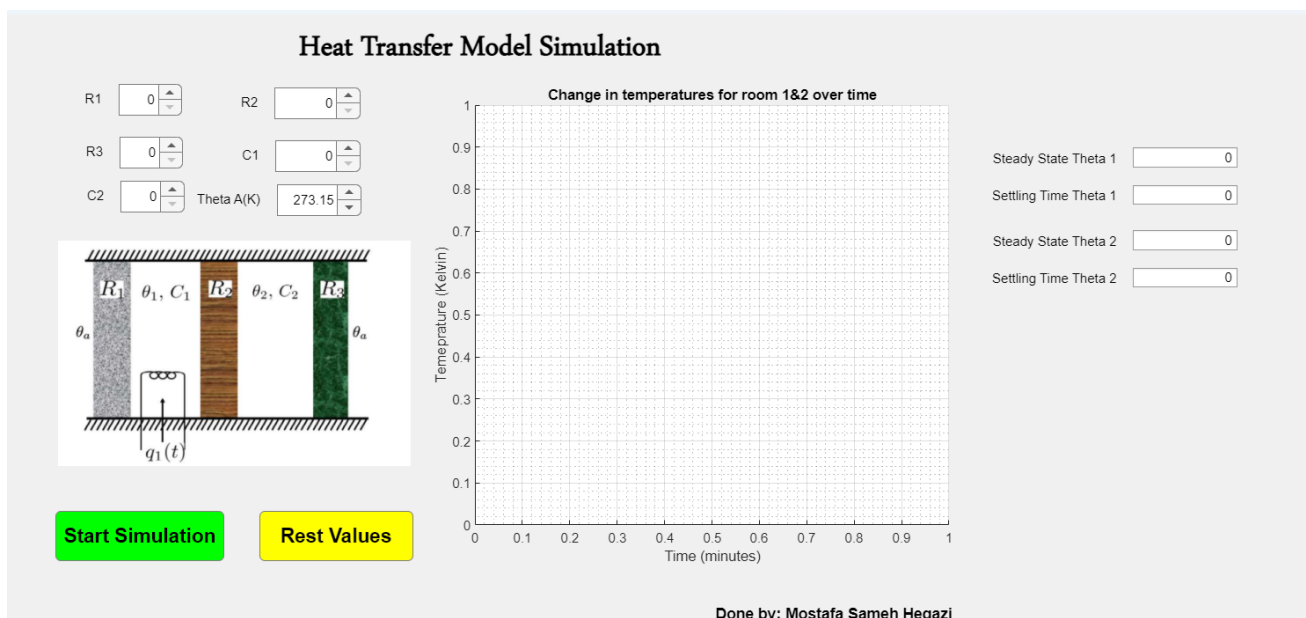
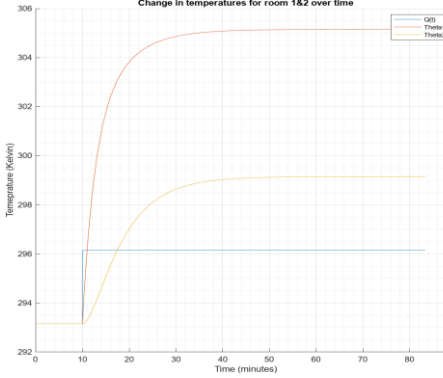
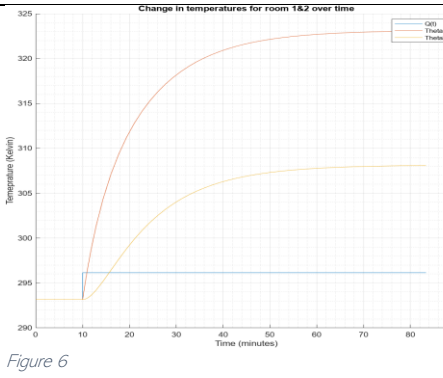
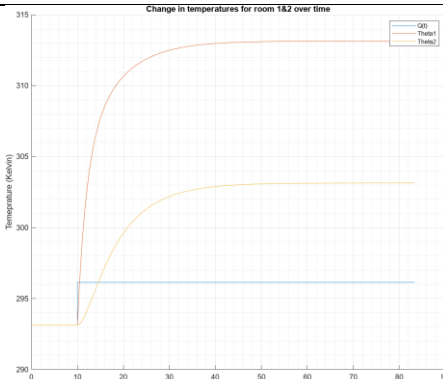
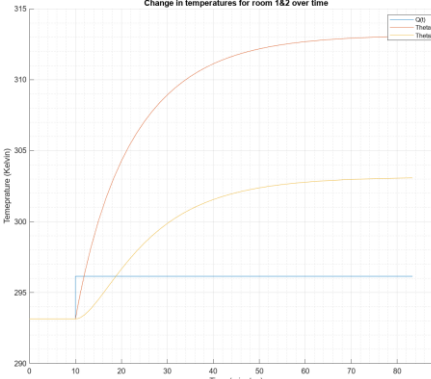


Figure 3

Procedure:

By taking each parameter individually, we change the value of this parameter and keep the rest to values given in the previous section.

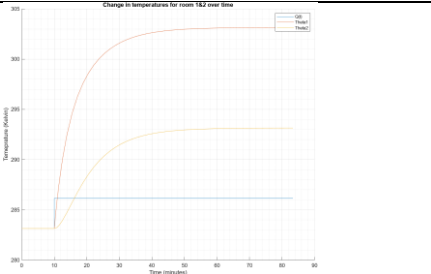
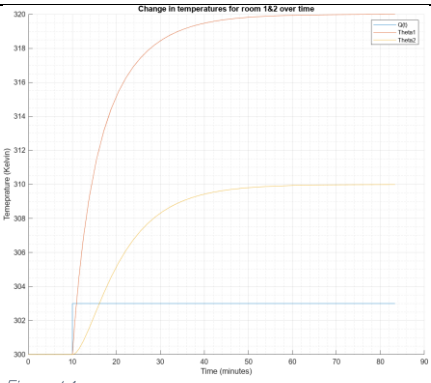
Considering R1			
New Value	Plot	Step-response (stepinfo)	Analysis
R1 = 5	 Figure 4	Steady State Theta 1 305 Settling Time Theta 1 24.62 Steady State Theta 2 299.1 Settling Time Theta 2 29.93 Figure 5	Both θ_1 and θ_2 have steady state values lower than table 1 since the value of R1 decreased, therefore more heat dissipated to environment. As R1 is halved, the values of settling time decreased from the values in table 1 for both θ_1 and θ_2 since the steady state decreased, therefore it takes less time for the system to settle and be stable.
R1 = 20	 Figure 6	Steady State Theta 1 323 Settling Time Theta 1 49.19 Steady State Theta 2 308.1 Settling Time Theta 2 51.22 Figure 7	Both θ_1 and θ_2 have higher steady state values since the value of R1 increased, therefore less heat dissipated to environment. As R1 is doubles, the value of the settling time for θ_1 and θ_2 increases over the values in table 1, since the steady state is higher which will take more time for the system to settle.
Conclusion: Depending on what have been observed, R1 have a directly proportional effect on the way Theta1 and Theta2 transfer functions behaves. Since it controls the amount of heat lost from the system.			
Considering C1			
New value	Plot	Step-response (stepinfo)	Analysis
C1 = 25	 Figure 8	Steady State Theta 1 313 Settling Time Theta 1 28.53 Steady State Theta 2 303.1 Settling Time Theta 2 33.56 Figure 9	In θ_1 and θ_2 settling time decreased compared to the values in Table 1. As the room capacity decreased making it faster for the temperature to settle. The steady state (peak value) did not change from the initial analysis in Table 1 and Figure 2. This is due to heat loss due to resistances and heat dissipation is the same, steady state value did not change.

$C1 = 100$	 Figure 10	Steady State Theta 1 313 Settling Time Theta 1 51.41 Steady State Theta 2 303.1 Settling Time Theta 2 51.99 Figure 11	In θ_1 and θ_2 settling time increased compared to the values in Table 1. As the room capacity increased making it slower for the temperature to settle. The steady state (peak value) did not change from the initial analysis in Table 1 and Figure 2. This is due to heat dissipation is the same, steady state value did not change.
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Conclusion:

The change in $C1$ does not affect the steady state value, however it affects the time taken for the system to respond. This is due to the time taken for the room temperature to get saturated will vary according to room capacitance. Therefore -as analysed above- as $C1$ increases, settling time proportionally increases.

Considering θ_a

New value	Plot	Step-response (stepinfo)	Analysis
$\theta_a = 283.15$	 Figure 12	Steady State Theta 1 303 Settling Time Theta 1 38.67 Steady State Theta 2 293.1 Settling Time Theta 2 43.2 Figure 13	When θ_a decreased by 10°K, the steady state for θ_1 and θ_2 decreased with the same value. Since the θ_a effects the initial values for θ_1 and θ_2
$\theta_a = 300$	 Figure 14	Steady State Theta 1 320 Settling Time Theta 1 34.63 Steady State Theta 2 310 Settling Time Theta 2 37.62 Figure 15	When θ_a increased by 10°K, the steady state for θ_1 and θ_2 increased with the same value. Since the θ_a effects the initial values for θ_1 and θ_2

Conclusion:

The value of θ_a affects directly the initial values of θ_1 and θ_2 which causes the difference in the steady-state value.